```
-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive: CoLaus x 20 imputed datasets ......................................
-> 45 targets ...............                [31.5s, 3+, 19-]
[observed/20] deriving ...


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6719 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20404.
! 27 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 18833 (92.1%)
. sumalco present (comparison source) : 11045 (54%)
. alcohol_category_conso derived : 18833 (92.1%)
. alcohol_category_sumalco derived: 11045 (54%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5161 27.4 2 Light 4107 21.8 3 Moderate 4192 22.3 4 Heavy 5373 28.5 5 NA 1609
8.5
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2014 18.2 2 Light 469 4.2 3 Moderate 866 7.8 4 Heavy 7696 69.7 5 NA
9397 85.1
. Source agreement (rows where both present, n = 10726):
. Agree : 4811 (44.9%)
. Conso higher : 336 (3.1%)
. Sumalco higher: 5579 (52%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 259)
-> 45 targets ...............                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 284)
-> 45 targets ...............                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 512)
-> 45 targets ...............                [31.5s, 3+, 19-]
```

```
. conso: Light / sumalco: Non−drinker (n = 160)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 457)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 1755)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 71)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 21)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2312)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 61)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 7)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 16)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 53 (0.3%)
. Sources available (1/2/3): 670 / 11725 / 7994
. Disagreement between any sources: 4.2%
. Disagreement between FPG and HbA1c: 1.9%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20422 | missing: 20
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17439 rows
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 9516 | missing (esthrp NA): 10926
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 5840
61.4 2 Past HRT 3 0 3 Current HRT 3673 38.6 4 NA 10926 115.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4089):
  Low < 168 | Medium 168−283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4089)
  Low 1357 (33.2%) Medium 1368 (33.5%) High 1364 (33.4%)
WHO corrected (N = 4089)
  Low 1419 (34.7%) Medium 1736 (42.5%) High 934 (22.8%)
. Visit: F2
Tertile (N = 2629)
  Low 869 (33.1%) Medium 928 (35.3%) High 832 (31.6%)
WHO corrected (N = 2629)
  Low 931 (35.4%) Medium 1153 (43.9%) High 545 (20.7%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Method alignment (tertile vs WHO corrected)
−> 45 targets ................                 [31.5s, 3+, 19−]
. Visit F1: 3397/4089 in same category (83.1%)
. Visit F2: 2172/2629 in same category (82.6%)


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Dairy sub−category derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_dairy: derived variables added.
  total: 10696
  fermented: 10795
  non−fermented: 10736
  low−fat: 10755
  high−fat: 10704
  total (excl. fondue): 10699
! .visit is an unordered factor. Using current factor level order.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Cumulative average dairy derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 12502
−> 45 targets ................                [31.5s, 3+, 19−]

  dairy_fermented_gday_cumavg: 12516
−> 45 targets ................                [31.5s, 3+, 19−]

  dairy_non_fermented_gday_cumavg: 12504
−> 45 targets ................                [31.5s, 3+, 19−]

  dairy_lowfat_gday_cumavg: 12511
−> 45 targets ................                [31.5s, 3+, 19−]

  dairy_highfat_gday_cumavg: 12502
−> 45 targets ................                [31.5s, 3+, 19−]

  dairy_total_nofondue_gday_cumavg: 12502
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Per−item dairy FFQ cumulative average derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

  FFQ1amount_cumavg: 12538
−> 45 targets ................                [31.5s, 3+, 19−]

  FFQ2amount_cumavg: 12524
−> 45 targets ................                [31.5s, 3+, 19−]
```

```
  FFQ3amount_cumavg: 12533
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 12528
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 12527
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 12534
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 12543
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 12528
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 12524
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 12513
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 12523
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 12525
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 12516
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 12524
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 12522
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 12523
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 12521
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 12524
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 12502
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 12569
. prot_content_nondairy_cumavg valid: 12500


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 11027 (53.9%)
. dairy_portion_total derived : 11027 (53.9%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 8634 42.2 2 >= 2 servings/day 2393 11.7 3 NA 9415 46.1
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 9887 48.4 2 >= 2 servings/day 1140 5.6 3 NA 9415 46.1
. Method agreement (rows with both non-missing, n = 11027):
. Agree : 9774 (88.6%)
. Disagree: 1253 (11.4%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1253
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4605 472 10.2 2 F2 3719 465 12.5 3 F3 2703 316 11.7


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Quartiles ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Baseline quartile cut-points (g/day, n = 4605):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 196.8
. Q3 | Q4 boundary : 302.2
. Overall quartile cut-points (g/day, n = 12502):
. Q1 | Q2 boundary : 116.5
. Q2 | Q3 boundary : 195
. Q3 | Q4 boundary : 295.8
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
```

```
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1152 22.7  7 F1 Q2 1151 22.7  8 F1 Q3 1151 22.7  9 F1
Q4 1151 22.7 10 F1 NA 459 9.1 11 F2 Q1 979 20 12 F2 Q2 1202 24.6 13 F2 Q3 1137
23.2 14 F2 Q4 1037 21.2 15 F2 NA 539 11 16 F3 Q1 755 20.1 17 F3 Q2 1073 28.6 18
F3 Q3 906 24.2 19 F3 Q4 808 21.5 20 F3 NA 209 5.6
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1231 24.3  7 F1 Q2 1048 20.7  8 F1 Q3 1121 22.1  9 F1
Q4 1205 23.8 10 F1 NA 459 9.1 11 F2 Q1 1067 21.8 12 F2 Q2 1089 22.3 13 F2 Q3
1119 22.9 14 F2 Q4 1080 22.1 15 F2 NA 539 11 16 F3 Q1 829 22.1 17 F3 Q2 987
26.3 18 F3 Q3 885 23.6 19 F3 Q4 841 22.4 20 F3 NA 209 5.6
. Dairy quartiles derived.
. dairy_quartile_baseline: 12502 / 20442 rows non−missing
. dairy_quartile_overall : 12502 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  12502):
. Same quartile : 12070 (96.5%)
. Different quartile: 432 (3.5%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 241 2 Q2 Q3 61 3 Q3 Q4 130
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4605 157 3.4 2 F2 4355 156 3.6 3 F3 3542 119 3.4


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 18926 2
derived 698 3 NA 818


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 19759 | missing: 683
```

```
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.1 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ...............                 [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ...............                 [31.5s, 3+, 19−]


−> 45 targets ...............                 [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (683 missing BMI).
  Prevalence of Obesity: 17.4%
  Prevalence of Overweight + Obesity: 55.7%


−> 45 targets ...............                 [31.5s, 3+, 19−]

.. Derive Age ..
−> 45 targets ...............                 [31.5s, 3+, 19−]


−> 45 targets ...............                 [31.5s, 3+, 19−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ...............                 [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ...............                 [31.5s, 3+, 19−]


−> 45 targets ...............                 [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810
```

```
−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8720 | No: 11709 | Missing: 13
. Prevalence: 42.7% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8720
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[1/20] deriving ...


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5563 27.7 2 Light 4401 21.9 3 Moderate 4455 22.1 4 Heavy 5700 28.3 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2437 18.6 2 Light 567 4.3 3 Moderate 1028 7.8 4 Heavy 9077 69.2 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5731 (43.8%)
. Conso higher : 514 (3.9%)
. Sumalco higher: 6832 (52.2%)
Breakdown of mismatches:
```

```
. conso: Non−drinker / sumalco: Light (n = 327)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 390)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 811)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 271)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 514)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2060)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 100)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 34)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2730)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 80)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 11)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 18)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive CVD Status ..
```

```
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2994 rows | 1033 / 6746 participants
. No : 17432 rows


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10264 56.5 3 Current HRT 6770 37.3 4 NA 2268 12.5


-> 45 targets ................                [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 168 | Medium 168-284 | High >= 284 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1610 (33.1%) Medium 1625 (33.4%) High 1632 (33.5%)
WHO corrected (N = 4867)
  Low 1698 (34.9%) Medium 2049 (42.1%) High 1120 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1454 (32.5%) Medium 1589 (35.6%) High 1424 (31.9%)
WHO corrected (N = 4467)
  Low 1530 (34.3%) Medium 1970 (44.1%) High 967 (21.6%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4031/4867 in same category (82.8%)
. Visit F2: 3688/4467 in same category (82.6%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
−> 45 targets ................                [31.5s, 3+, 19−]

.. Dairy protein content derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


−> 45 targets ................                [31.5s, 3+, 19−]

.. Non−dairy protein cumulative average derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10218 50 2 >= 2 servings/day 2890 14.1 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11567 56.6 2 >= 2 servings/day 1541 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11759 (89.7%)
. Disagree: 1349 (10.3%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1349
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 495 10 2 F2 4582 501 10.9 3 F3 3575 353 9.9


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
```

```
. Q1 | Q2 boundary : 109
. Q2 | Q3 boundary : 195.9
. Q3 | Q4 boundary : 302.2
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.6
. Q2 | Q3 boundary : 198.7
. Q3 | Q4 boundary : 295.5
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 844 17.2 12 F2 Q2 1343 27.4 13 F2 Q3
1277 26.1 14 F2 Q4 1096 22.4 15 F2 NA 334 6.8 16 F3 Q1 606 16.2 17 F3 Q2 1176
31.4 18 F3 Q3 1061 28.3 19 F3 Q4 799 21.3 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1443 28.5  7 F1 Q2 1077 21.3  8 F1 Q3 1134 22.4  9 F1
Q4 1297 25.6 10 F1 NA 113 2.2 11 F2 Q1 1068 21.8 12 F2 Q2 1159 23.7 13 F2 Q3
1187 24.3 14 F2 Q4 1146 23.4 15 F2 NA 334 6.8 16 F3 Q1 778 20.7 17 F3 Q2 1052
28 18 F3 Q3 967 25.8 19 F3 Q4 845 22.5 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12265 (93.2%)
. Different quartile: 888 (6.8%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 601 2 Q3 Q2 132 3 Q3 Q4 155
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 308 6.2 2 F2 4560 314 6.9 3 F3 3642 266 7.3


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19494 2
derived 778 3 NA 170
```

```
-> 45 targets ................              [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................              [31.5s, 3+, 19-]


-> 45 targets ................              [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.1 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................              [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................              [31.5s, 3+, 19-]


-> 45 targets ................              [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.4%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ................              [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................              [31.5s, 3+, 19-]


-> 45 targets ................              [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................              [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................              [31.5s, 3+, 19-]


-> 45 targets ................              [31.5s, 3+, 19-]
```

```
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8774 | No: 11655 | Missing: 13
. Prevalence: 42.9% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8774
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[2/20] deriving ...


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5508 27.4 2 Light 4391 21.8 3 Moderate 4503 22.4 4 Heavy 5717 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
```

```
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2417 18.4 2 Light 591 4.5 3 Moderate 1060 8.1 4 Heavy 9041 69 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5728 (43.8%)
. Conso higher : 535 (4.1%)
. Sumalco higher: 6814 (52.1%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 342)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 394)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 773)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 270)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 525)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2031)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 116)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 35)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2749)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 81)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 10)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 23)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
```

```
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10334 56.9 3 Current HRT 6700 36.9 4 NA 2268 12.5


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 166 | Medium 166−281 | High >= 281 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1609 (33.1%) Medium 1630 (33.5%) High 1628 (33.4%)
WHO corrected (N = 4867)
  Low 1712 (35.2%) Medium 2060 (42.3%) High 1095 (22.5%)
. Visit: F2
Tertile (N = 4467)
  Low 1416 (31.7%) Medium 1590 (35.6%) High 1461 (32.7%)
WHO corrected (N = 4467)
```

```
  Low 1557 (34.9%) Medium 1939 (43.4%) High 971 (21.7%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 3997/4867 in same category (82.1%)
. Visit F2: 3662/4467 in same category (82.0%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  dairy_total_nofondue_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Per−item dairy FFQ cumulative average derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ1amount_cumavg: 13170
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ2amount_cumavg: 13162
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ3amount_cumavg: 13167
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ4amount_cumavg: 13166
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ5amount_cumavg: 13163
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ6amount_cumavg: 13167
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ7amount_cumavg: 13174
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ8amount_cumavg: 13164
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ52amount_cumavg: 13169
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ53amount_cumavg: 13161
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ63amount_cumavg: 13163
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ68amount_cumavg: 13164
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ71amount_cumavg: 13161
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ82amount_cumavg: 13160
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ83amount_cumavg: 13159
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ84amount_cumavg: 13162
```

```
−> 45 targets ................                [31.5s, 3+, 19−]

  FFQ85amount_cumavg: 13160
−> 45 targets ................                [31.5s, 3+, 19−]

  FFQ86amount_cumavg: 13163
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Dairy protein content derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


−> 45 targets ................                [31.5s, 3+, 19−]

.. Non−dairy protein cumulative average derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10222 50 2 >= 2 servings/day 2886 14.1 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11566 56.6 2 >= 2 servings/day 1542 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11764 (89.7%)
. Disagree: 1344 (10.3%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1344
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 487 9.8 2 F2 4582 494 10.8 3 F3 3575 363 10.2
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Quartiles ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Baseline quartile cut-points (g/day, n = 4951):
. Q1 | Q2 boundary : 110
. Q2 | Q3 boundary : 196.3
. Q3 | Q4 boundary : 301.3
. Overall quartile cut-points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.6
. Q2 | Q3 boundary : 198.8
. Q3 | Q4 boundary : 293.8
. Distribution of 'dairy_quartile_baseline' by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 854 17.4 12 F2 Q2 1327 27.1 13 F2 Q3
1290 26.4 14 F2 Q4 1089 22.3 15 F2 NA 334 6.8 16 F3 Q1 644 17.2 17 F3 Q2 1157
30.8 18 F3 Q3 1067 28.4 19 F3 Q4 774 20.6 20 F3 NA 109 2.9
. Distribution of 'dairy_quartile_overall' by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1427 28.2  7 F1 Q2 1085 21.4  8 F1 Q3 1125 22.2  9 F1
Q4 1314 25.9 10 F1 NA 113 2.2 11 F2 Q1 1061 21.7 12 F2 Q2 1160 23.7 13 F2 Q3
1191 24.3 14 F2 Q4 1148 23.5 15 F2 NA 334 6.8 16 F3 Q1 801 21.4 17 F3 Q2 1043
27.8 18 F3 Q3 972 25.9 19 F3 Q4 826 22 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non-missing
. dairy_quartile_overall : 13153 / 20442 rows non-missing
. Baseline vs overall quartile agreement (rows with both non-missing, n =
  13153):
. Same quartile : 12294 (93.5%)
. Different quartile: 859 (6.5%)
Cross-tabulation of differing classifications (baseline -> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 553 2 Q3 Q2 119 3 Q3 Q4 187
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 301 6.1 2 F2 4560 306 6.7 3 F3 3642 252 6.9


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive ATCs ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_atc: ATC-based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
.. Derive Smoking Status ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19513 2
derived 759 3 NA 170


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8784 | No: 11645 | Missing: 13
. Prevalence: 43% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8784
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[3/20] deriving ...


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5532 27.5 2 Light 4369 21.7 3 Moderate 4509 22.4 4 Heavy 5709 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2443 18.6 2 Light 548 4.2 3 Moderate 1042 7.9 4 Heavy 9076 69.2 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5763 (44.1%)
. Conso higher : 513 (3.9%)
. Sumalco higher: 6801 (52%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 304)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 384)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 801)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 256)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 522)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2029)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 115)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 33)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2761)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 78)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 10)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 21)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
```

```
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self-report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2990 rows | 1032 / 6746 participants
. No : 17436 rows


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10315 56.8 3 Current HRT 6719 37 4 NA 2268 12.5


-> 45 targets ................                [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 169 | Medium 169-286 | High >= 286 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
```

```
. Visit: F1
Tertile (N = 4867)
  Low 1605 (33.0%) Medium 1634 (33.6%) High 1628 (33.4%)
WHO corrected (N = 4867)
  Low 1660 (34.1%) Medium 2076 (42.7%) High 1131 (23.2%)
. Visit: F2
Tertile (N = 4467)
  Low 1462 (32.7%) Medium 1594 (35.7%) High 1411 (31.6%)
WHO corrected (N = 4467)
  Low 1535 (34.4%) Medium 1948 (43.6%) High 984 (22.0%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4061/4867 in same category (83.4%)
. Visit F2: 3759/4467 in same category (84.2%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10234 50.1 2 >= 2 servings/day 2874 14.1 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11592 56.7 2 >= 2 servings/day 1516 7.4 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
```

. Agree : 11750 (89.6%)
. Disagree: 1358 (10.4%)
Cross−tabulation of disagreements (freq −> portion):
A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1358
Disagreement rate by visit:
A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 499 10.1 2 F2 4582 499 10.9 3 F3 3575 360 10.1

−> 45 targets .... [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets .... [31.5s, 3+, 19−]

−> 45 targets .... [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 109.4
. Q2 | Q3 boundary : 195.6
. Q3 | Q4 boundary : 300.3
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.7
. Q2 | Q3 boundary : 198
. Q3 | Q4 boundary : 293.6
. Distribution of ‘dairy_quartile_baseline‘ by visit:
A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0 2 Baseline Q2 0 0 3 Baseline Q3 0 0 4 Baseline Q4 0 0 5
Baseline NA 6733 100 6 F1 Q1 1238 24.4 7 F1 Q2 1238 24.4 8 F1 Q3 1237 24.4 9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 857 17.5 12 F2 Q2 1347 27.5 13 F2 Q3
1274 26 14 F2 Q4 1082 22.1 15 F2 NA 334 6.8 16 F3 Q1 611 16.3 17 F3 Q2 1194
31.8 18 F3 Q3 1028 27.4 19 F3 Q4 809 21.6 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0 2 Baseline Q2 0 0 3 Baseline Q3 0 0 4 Baseline Q4 0 0 5
Baseline NA 6733 100 6 F1 Q1 1444 28.5 7 F1 Q2 1064 21 8 F1 Q3 1143 22.6 9 F1
Q4 1300 25.7 10 F1 NA 113 2.2 11 F2 Q1 1070 21.9 12 F2 Q2 1170 23.9 13 F2 Q3
1181 24.1 14 F2 Q4 1139 23.3 15 F2 NA 334 6.8 16 F3 Q1 775 20.7 17 F3 Q2 1054
28.1 18 F3 Q3 964 25.7 19 F3 Q4 849 22.6 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
13153):
. Same quartile : 12319 (93.7%)
. Different quartile: 834 (6.3%)
Cross−tabulation of differing classifications (baseline −> overall):
A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 583 2 Q3 Q2 92 3 Q3 Q4 159
Disagreement rate by visit:
A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 300 6.1 2 F2 4560 306 6.7 3 F3 3642 228 6.3

−> 45 targets .... [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets .... [31.5s, 3+, 19−]

```
−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19494 2
derived 778 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.4 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.9%


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Age ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
```

```
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ................                [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8774 | No: 11655 | Missing: 13
. Prevalence: 42.9% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8774
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[4/20] deriving ...


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5570 27.7 2 Light 4366 21.7 3 Moderate 4481 22.3 4 Heavy 5702 28.3 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2435 18.6 2 Light 558 4.3 3 Moderate 1061 8.1 4 Heavy 9055 69.1 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5770 (44.1%)
. Conso higher : 488 (3.7%)
. Sumalco higher: 6819 (52.1%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 316)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 405)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 794)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 247)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 523)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2033)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 103)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 31)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2748)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 79)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 9)
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 19)
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Diabetes Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self-report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10275 56.5 3 Current HRT 6759 37.2 4 NA 2268 12.5


-> 45 targets ................                [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
. Tertile cut−points (from F1, N = 4867):
  Low < 165 | Medium 165−282 | High >= 282 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1610 (33.1%) Medium 1629 (33.5%) High 1628 (33.4%)
WHO corrected (N = 4867)
  Low 1719 (35.3%) Medium 2027 (41.6%) High 1121 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1431 (32.0%) Medium 1612 (36.1%) High 1424 (31.9%)
WHO corrected (N = 4467)
  Low 1540 (34.5%) Medium 1973 (44.2%) High 954 (21.4%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Method alignment (tertile vs WHO corrected)
−> 45 targets ................                 [31.5s, 3+, 19−]
. Visit F1: 4029/4867 in same category (82.8%)
. Visit F2: 3658/4467 in same category (81.9%)


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Dairy sub−category derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non−fermented: 13077
  low−fat: 13080
  high−fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Cumulative average dairy derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
```

```
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10249 50.1 2 >= 2 servings/day 2859 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11585 56.7 2 >= 2 servings/day 1523 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11772 (89.8%)
. Disagree: 1336 (10.2%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1336
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 501 10.1 2 F2 4582 487 10.6 3 F3 3575 348 9.7


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 197
. Q3 | Q4 boundary : 302.1
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.7
. Q2 | Q3 boundary : 198.9
. Q3 | Q4 boundary : 294.6
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1238 24.4  9 F1
Q4 1237 24.4 10 F1 NA 113 2.2 11 F2 Q1 846 17.3 12 F2 Q2 1361 27.8 13 F2 Q3
1275 26.1 14 F2 Q4 1078 22 15 F2 NA 334 6.8 16 F3 Q1 610 16.3 17 F3 Q2 1199 32
18 F3 Q3 1044 27.8 19 F3 Q4 789 21 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1444 28.5  7 F1 Q2 1057 20.9  8 F1 Q3 1137 22.5  9 F1
Q4 1313 25.9 10 F1 NA 113 2.2 11 F2 Q1 1054 21.5 12 F2 Q2 1186 24.2 13 F2 Q3
1188 24.3 14 F2 Q4 1132 23.1 15 F2 NA 334 6.8 16 F3 Q1 791 21.1 17 F3 Q2 1045
27.9 18 F3 Q3 963 25.7 19 F3 Q4 843 22.5 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12289 (93.4%)
. Different quartile: 864 (6.6%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 595 2 Q3 Q2 85 3 Q3 Q4 184
```

```
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 307 6.2 2 F2 4560 295 6.5 3 F3 3642 262 7.2


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive ATCs ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_atc: ATC-based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Smoking Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19502 2
derived 770 3 NA 170


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.3 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.9%
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8774 | No: 11655 | Missing: 13
. Prevalence: 42.9% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8774
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[5/20] deriving ...


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Education Level ..
```

```
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5536 27.5 2 Light 4372 21.7 3 Moderate 4508 22.4 4 Heavy 5703 28.3 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2409 18.4 2 Light 570 4.3 3 Moderate 1038 7.9 4 Heavy 9092 69.4 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5732 (43.8%)
. Conso higher : 500 (3.8%)
. Sumalco higher: 6845 (52.3%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 331)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 392)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 793)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 257)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 507)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2047)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 103)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 29)
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2775)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 78)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 10)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 23)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10275 56.5 3 Current HRT 6759 37.2 4 NA 2268 12.5
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 168 | Medium 168-283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1611 (33.1%) Medium 1628 (33.4%) High 1628 (33.4%)
WHO corrected (N = 4867)
  Low 1680 (34.5%) Medium 2066 (42.4%) High 1121 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1462 (32.7%) Medium 1572 (35.2%) High 1433 (32.1%)
WHO corrected (N = 4467)
  Low 1547 (34.6%) Medium 1923 (43.0%) High 997 (22.3%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4051/4867 in same category (83.2%)
. Visit F2: 3714/4467 in same category (83.1%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Cumulative average dairy derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_fermented_gday_cumavg: 13156
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_non_fermented_gday_cumavg: 13154
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_lowfat_gday_cumavg: 13156
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_highfat_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_total_nofondue_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Per−item dairy FFQ cumulative average derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ1amount_cumavg: 13170
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ2amount_cumavg: 13162
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ3amount_cumavg: 13167
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ4amount_cumavg: 13166
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ5amount_cumavg: 13163
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ6amount_cumavg: 13167
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ7amount_cumavg: 13174
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10283 50.3 2 >= 2 servings/day 2825 13.8 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11598 56.7 2 >= 2 servings/day 1510 7.4 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
. Agree : 11793 (90%)
. Disagree: 1315 (10%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1315
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 491 9.9 2 F2 4582 487 10.6 3 F3 3575 337 9.4


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Quartiles ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Baseline quartile cut-points (g/day, n = 4951):
. Q1 | Q2 boundary : 109.6
. Q2 | Q3 boundary : 197.5
. Q3 | Q4 boundary : 302.8
. Overall quartile cut-points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.5
. Q2 | Q3 boundary : 198.7
. Q3 | Q4 boundary : 294.9
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 850 17.4 12 F2 Q2 1368 28 13 F2 Q3 1264
25.8 14 F2 Q4 1078 22 15 F2 NA 334 6.8 16 F3 Q1 610 16.3 17 F3 Q2 1218 32.5 18
F3 Q3 1046 27.9 19 F3 Q4 768 20.5 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1443 28.5  7 F1 Q2 1050 20.7  8 F1 Q3 1137 22.5  9 F1
Q4 1321 26.1 10 F1 NA 113 2.2 11 F2 Q1 1058 21.6 12 F2 Q2 1176 24 13 F2 Q3 1182
24.2 14 F2 Q4 1144 23.4 15 F2 NA 334 6.8 16 F3 Q1 788 21 17 F3 Q2 1062 28.3 18
F3 Q3 969 25.8 19 F3 Q4 823 21.9 20 F3 NA 109 2.9
. Dairy quartiles derived.
```

```
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12303 (93.5%)
. Different quartile: 850 (6.5%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 591 2 Q3 Q2 55 3 Q3 Q4 204
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 305 6.2 2 F2 4560 290 6.4 3 F3 3642 255 7


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19496 2
derived 776 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Deriving BMI category ..
```

```
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8771 | No: 11658 | Missing: 13
. Prevalence: 42.9% (among non-missing rows)
. Source driving 'Yes' classification:
```

```
. HTA (measured BP, tier 1) : 8771
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[6/20] deriving ...


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5540 27.5 2 Light 4397 21.9 3 Moderate 4471 22.2 4 Heavy 5711 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2397 18.3 2 Light 557 4.2 3 Moderate 1036 7.9 4 Heavy 9119 69.6 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5709 (43.7%)
. Conso higher : 499 (3.8%)
. Sumalco higher: 6869 (52.5%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 329)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 387)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 823)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 256)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 518)
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2068)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 105)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 26)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2744)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 84)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 8)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 20)
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Diabetes Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2.1%
. Self-report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive HRT Status ..
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10257 56.4 3 Current HRT 6777 37.3 4 NA 2268 12.5


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 169 | Medium 169-284 | High >= 284 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1618 (33.2%) Medium 1621 (33.3%) High 1628 (33.4%)
WHO corrected (N = 4867)
  Low 1667 (34.3%) Medium 2081 (42.8%) High 1119 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1476 (33.0%) Medium 1593 (35.7%) High 1398 (31.3%)
WHO corrected (N = 4467)
  Low 1551 (34.7%) Medium 1970 (44.1%) High 946 (21.2%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4037/4867 in same category (82.9%)
. Visit F2: 3722/4467 in same category (83.3%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
.. Non−dairy protein cumulative average derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10261 50.2 2 >= 2 servings/day 2847 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11595 56.7 2 >= 2 servings/day 1513 7.4 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11774 (89.8%)
. Disagree: 1334 (10.2%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1334
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 497 10 2 F2 4582 485 10.6 3 F3 3575 352 9.8


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 109.8
. Q2 | Q3 boundary : 195.9
. Q3 | Q4 boundary : 300.8
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123
. Q2 | Q3 boundary : 198.4
. Q3 | Q4 boundary : 293.9
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 850 17.4 12 F2 Q2 1347 27.5 13 F2 Q3
1279 26.1 14 F2 Q4 1084 22.1 15 F2 NA 334 6.8 16 F3 Q1 614 16.4 17 F3 Q2 1185
```

```
31.6 18 F3 Q3 1047 27.9 19 F3 Q4 796 21.2 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1440 28.4  7 F1 Q2 1070 21.1  8 F1 Q3 1134 22.4  9 F1
Q4 1307 25.8 10 F1 NA 113 2.2 11 F2 Q1 1069 21.8 12 F2 Q2 1162 23.7 13 F2 Q3
1196 24.4 14 F2 Q4 1133 23.2 15 F2 NA 334 6.8 16 F3 Q1 780 20.8 17 F3 Q2 1056
28.2 18 F3 Q3 958 25.5 19 F3 Q4 848 22.6 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12291 (93.4%)
. Different quartile: 862 (6.6%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 587 2 Q3 Q2 105 3 Q3 Q4 170
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 305 6.2 2 F2 4560 302 6.6 3 F3 3642 255 7


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19506 2
derived 766 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
```

```
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ...............                 [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ...............                 [31.5s, 3+, 19−]


−> 45 targets ...............                 [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.9%


−> 45 targets ...............                 [31.5s, 3+, 19−]

.. Derive Age ..
−> 45 targets ...............                 [31.5s, 3+, 19−]


−> 45 targets ...............                 [31.5s, 3+, 19−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ...............                 [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ...............                 [31.5s, 3+, 19−]


−> 45 targets ...............                 [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ...............                 [31.5s, 3+, 19−]
```

```
.. Derive htn Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8774 | No: 11655 | Missing: 13
. Prevalence: 42.9% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8774
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[7/20] deriving ...


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5549 27.6 2 Light 4383 21.8 3 Moderate 4474 22.2 4 Heavy 5713 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2454 18.7 2 Light 574 4.4 3 Moderate 1042 7.9 4 Heavy 9039 69 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5733 (43.8%)
. Conso higher : 544 (4.2%)
. Sumalco higher: 6800 (52%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 332)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 399)
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 792)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 282)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 509)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2033)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 106)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 37)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2735)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 89)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 7)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 23)
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Diabetes Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self-report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10241 56.3 3 Current HRT 6793 37.4 4 NA 2268 12.5


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 169 | Medium 169-283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1617 (33.2%) Medium 1624 (33.4%) High 1626 (33.4%)
WHO corrected (N = 4867)
  Low 1682 (34.6%) Medium 2085 (42.8%) High 1100 (22.6%)
. Visit: F2
Tertile (N = 4467)
  Low 1503 (33.6%) Medium 1530 (34.3%) High 1434 (32.1%)
WHO corrected (N = 4467)
  Low 1561 (34.9%) Medium 1939 (43.4%) High 967 (21.6%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4032/4867 in same category (82.8%)
```

```
. Visit F2: 3710/4467 in same category (83.1%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
−> 45 targets ................                [31.5s, 3+, 19−]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


−> 45 targets ................                [31.5s, 3+, 19−]

.. Non−dairy protein cumulative average derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10247 50.1 2 >= 2 servings/day 2861 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11576 56.6 2 >= 2 servings/day 1532 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11779 (89.9%)
. Disagree: 1329 (10.1%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1329
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 493 10 2 F2 4582 489 10.7 3 F3 3575 347 9.7


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 196.1
. Q3 | Q4 boundary : 301.7
. Overall quartile cut−points (g/day, n = 13153):
```

```
. Q1 | Q2 boundary : 122.9
. Q2 | Q3 boundary : 197.7
. Q3 | Q4 boundary : 294.3
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 880 18 12 F2 Q2 1332 27.2 13 F2 Q3 1268
25.9 14 F2 Q4 1080 22.1 15 F2 NA 334 6.8 16 F3 Q1 638 17 17 F3 Q2 1185 31.6 18
F3 Q3 1030 27.5 19 F3 Q4 789 21 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1426 28.2  7 F1 Q2 1071 21.1  8 F1 Q3 1142 22.6  9 F1
Q4 1312 25.9 10 F1 NA 113 2.2 11 F2 Q1 1066 21.8 12 F2 Q2 1173 24 13 F2 Q3 1187
24.3 14 F2 Q4 1134 23.2 15 F2 NA 334 6.8 16 F3 Q1 797 21.2 17 F3 Q2 1044 27.8
18 F3 Q3 959 25.6 19 F3 Q4 842 22.4 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12373 (94.1%)
. Different quartile: 780 (5.9%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 533 2 Q3 Q2 66 3 Q3 Q4 181
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 283 5.7 2 F2 4560 267 5.9 3 F3 3642 230 6.3


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19496 2
derived 776 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
-> 45 targets ...............                 [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ...............                 [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ...............                 [31.5s, 3+, 19-]


-> 45 targets ...............                 [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ...............                 [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ...............                 [31.5s, 3+, 19-]


-> 45 targets ...............                 [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ...............                 [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ...............                 [31.5s, 3+, 19-]


-> 45 targets ...............                 [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
```

```
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810

−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]

−> 45 targets ................                 [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8774 | No: 11655 | Missing: 13
. Prevalence: 42.9% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8774
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[8/20] deriving ...

−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                 [31.5s, 3+, 19−]

−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).

−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                 [31.5s, 3+, 19−]

−> 45 targets ................                 [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5569 27.7 2 Light 4374 21.7 3 Moderate 4460 22.2 4 Heavy 5716 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2393 18.3 2 Light 591 4.5 3 Moderate 1021 7.8 4 Heavy 9104 69.4 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
```

```
. Agree : 5726 (43.8%)
. Conso higher : 499 (3.8%)
. Sumalco higher: 6852 (52.4%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 347)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 374)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 846)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 257)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 522)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2027)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 106)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 30)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2736)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 77)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 7)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 22)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2.1%
. Self−report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2986 rows | 1031 / 6746 participants
. No : 17440 rows


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10158 55.9 3 Current HRT 6876 37.8 4 NA 2268 12.5


-> 45 targets ................                [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 168 | Medium 168-282 | High >= 282 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1610 (33.1%) Medium 1623 (33.3%) High 1634 (33.6%)
WHO corrected (N = 4867)
  Low 1685 (34.6%) Medium 2064 (42.4%) High 1118 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1477 (33.1%) Medium 1583 (35.4%) High 1407 (31.5%)
WHO corrected (N = 4467)
  Low 1562 (35.0%) Medium 1928 (43.2%) High 977 (21.9%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
```

```
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                [31.5s, 3+, 19-]
. Visit F1: 4036/4867 in same category (82.9%)
. Visit F2: 3732/4467 in same category (83.5%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10240 50.1 2 >= 2 servings/day 2868 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11582 56.7 2 >= 2 servings/day 1526 7.5 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
. Agree : 11766 (89.8%)
. Disagree: 1342 (10.2%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1342
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 499 10.1 2 F2 4582 488 10.7 3 F3 3575 355 9.9


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Quartiles ..
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Baseline quartile cut-points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 196.1
. Q3 | Q4 boundary : 302.4
. Overall quartile cut-points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.5
. Q2 | Q3 boundary : 198.6
. Q3 | Q4 boundary : 294.7
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 897 18.3 12 F2 Q2 1290 26.4 13 F2 Q3
1293 26.4 14 F2 Q4 1080 22.1 15 F2 NA 334 6.8 16 F3 Q1 644 17.2 17 F3 Q2 1160
30.9 18 F3 Q3 1045 27.9 19 F3 Q4 793 21.1 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1431 28.3  7 F1 Q2 1082 21.4  8 F1 Q3 1125 22.2  9 F1
Q4 1313 25.9 10 F1 NA 113 2.2 11 F2 Q1 1072 21.9 12 F2 Q2 1157 23.6 13 F2 Q3
1195 24.4 14 F2 Q4 1136 23.2 15 F2 NA 334 6.8 16 F3 Q1 786 21 17 F3 Q2 1049 28
18 F3 Q3 968 25.8 19 F3 Q4 839 22.4 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non-missing
. dairy_quartile_overall : 13153 / 20442 rows non-missing
. Baseline vs overall quartile agreement (rows with both non-missing, n =
  13153):
. Same quartile : 12356 (93.9%)
. Different quartile: 797 (6.1%)
Cross-tabulation of differing classifications (baseline -> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 510 2 Q3 Q2 110 3 Q3 Q4 177
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 305 6.2 2 F2 4560 273 6 3 F3 3642 219 6


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive ATCs ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_atc: ATC-based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Smoking Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19481 2
derived 791 3 NA 170


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8773 | No: 11656 | Missing: 13
. Prevalence: 42.9% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8773
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[9/20] deriving ...


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
```

```
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5547 27.6 2 Light 4374 21.7 3 Moderate 4497 22.4 4 Heavy 5701 28.3 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2424 18.5 2 Light 577 4.4 3 Moderate 1055 8 4 Heavy 9053 69.1 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5711 (43.7%)
. Conso higher : 523 (4%)
. Sumalco higher: 6843 (52.3%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 334)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 397)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 801)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 266)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 530)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2018)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 109)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 33)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2763)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 83)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 11)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 21)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10223 56.3 3 Current HRT 6811 37.5 4 NA 2268 12.5


−> 45 targets ................                [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 168 | Medium 168−282 | High >= 282 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1610 (33.1%) Medium 1632 (33.5%) High 1625 (33.4%)
WHO corrected (N = 4867)
```

```
  Low 1683 (34.6%) Medium 2079 (42.7%) High 1105 (22.7%)
. Visit: F2
Tertile (N = 4467)
  Low 1482 (33.2%) Medium 1543 (34.5%) High 1442 (32.3%)
WHO corrected (N = 4467)
  Low 1566 (35.1%) Medium 1922 (43.0%) High 979 (21.9%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                [31.5s, 3+, 19-]
. Visit F1: 4036/4867 in same category (82.9%)
. Visit F2: 3688/4467 in same category (82.6%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10240 50.1 2 >= 2 servings/day 2868 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11583 56.7 2 >= 2 servings/day 1525 7.5 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
. Agree : 11765 (89.8%)
. Disagree: 1343 (10.2%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
```

```
<int> 1 >= 2 servings/day < 2 servings/day 1343
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 503 10.2 2 F2 4582 488 10.7 3 F3 3575 352 9.8


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 196.6
. Q3 | Q4 boundary : 303.3
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.3
. Q2 | Q3 boundary : 199.9
. Q3 | Q4 boundary : 295.5
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 871 17.8 12 F2 Q2 1314 26.8 13 F2 Q3
1289 26.3 14 F2 Q4 1086 22.2 15 F2 NA 334 6.8 16 F3 Q1 631 16.8 17 F3 Q2 1154
30.8 18 F3 Q3 1068 28.5 19 F3 Q4 789 21 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1437 28.4  7 F1 Q2 1081 21.3  8 F1 Q3 1127 22.3  9 F1
Q4 1306 25.8 10 F1 NA 113 2.2 11 F2 Q1 1062 21.7 12 F2 Q2 1162 23.7 13 F2 Q3
1189 24.3 14 F2 Q4 1147 23.4 15 F2 NA 334 6.8 16 F3 Q1 790 21.1 17 F3 Q2 1045
27.9 18 F3 Q3 972 25.9 19 F3 Q4 835 22.3 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12298 (93.5%)
. Different quartile: 855 (6.5%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 549 2 Q3 Q2 131 3 Q3 Q4 175
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 309 6.2 2 F2 4560 291 6.4 3 F3 3642 255 7


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
```

```
. Processed 20442 rows; 14079 rows had valid ATC mapping data.

-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Smoking Status ..
-> 45 targets ................                [31.5s, 3+, 19-]

-> 45 targets ................                [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19493 2
derived 779 3 NA 170

-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                [31.5s, 3+, 19-]

-> 45 targets ................                [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²

-> 45 targets ................                [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                [31.5s, 3+, 19-]

-> 45 targets ................                [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.9%

-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                [31.5s, 3+, 19-]

-> 45 targets ................                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
```

```
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8783 | No: 11646 | Missing: 13
. Prevalence: 43% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8783
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[10/20] deriving ...


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5562 27.6 2 Light 4371 21.7 3 Moderate 4486 22.3 4 Heavy 5700 28.3 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2385 18.2 2 Light 576 4.4 3 Moderate 1052 8 4 Heavy 9096 69.4 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5727 (43.8%)
. Conso higher : 476 (3.6%)
. Sumalco higher: 6874 (52.6%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 337)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 401)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 802)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 230)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 512)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2066)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 103)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 30)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2756)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 79)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 7)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 27)
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10155 55.9 3 Current HRT 6879 37.9 4 NA 2268 12.5


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 168 | Medium 168−283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
```

```
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1612 (33.1%) Medium 1623 (33.3%) High 1632 (33.5%)
WHO corrected (N = 4867)
  Low 1683 (34.6%) Medium 2073 (42.6%) High 1111 (22.8%)
. Visit: F2
Tertile (N = 4467)
  Low 1438 (32.2%) Medium 1558 (34.9%) High 1471 (32.9%)
WHO corrected (N = 4467)
  Low 1519 (34.0%) Medium 1959 (43.9%) High 989 (22.1%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4037/4867 in same category (82.9%)
. Visit F2: 3688/4467 in same category (82.6%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
  FFQ68amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
```

```
servings/day 10259 50.2 2 >= 2 servings/day 2849 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11583 56.7 2 >= 2 servings/day 1525 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11784 (89.9%)
. Disagree: 1324 (10.1%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1324
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 492 9.9 2 F2 4582 496 10.8 3 F3 3575 336 9.4


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.3
. Q2 | Q3 boundary : 196.4
. Q3 | Q4 boundary : 302.4
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.6
. Q2 | Q3 boundary : 198.9
. Q3 | Q4 boundary : 294.3
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 869 17.8 12 F2 Q2 1328 27.1 13 F2 Q3
1296 26.5 14 F2 Q4 1067 21.8 15 F2 NA 334 6.8 16 F3 Q1 644 17.2 17 F3 Q2 1150
30.7 18 F3 Q3 1067 28.4 19 F3 Q4 781 20.8 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1433 28.3  7 F1 Q2 1078 21.3  8 F1 Q3 1118 22.1  9 F1
Q4 1322 26.1 10 F1 NA 113 2.2 11 F2 Q1 1065 21.8 12 F2 Q2 1166 23.8 13 F2 Q3
1200 24.5 14 F2 Q4 1129 23.1 15 F2 NA 334 6.8 16 F3 Q1 791 21.1 17 F3 Q2 1044
27.8 18 F3 Q3 970 25.9 19 F3 Q4 837 22.3 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12303 (93.5%)
. Different quartile: 850 (6.5%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 538 2 Q3 Q2 110 3 Q3 Q4 202
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 314 6.3 2 F2 4560 292 6.4 3 F3 3642 244 6.7
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19500 2
derived 772 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.9%


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Age ..
```

```
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8776 | No: 11653 | Missing: 13
. Prevalence: 43% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8776
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[11/20] deriving ...


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5553 27.6 2 Light 4380 21.8 3 Moderate 4473 22.2 4 Heavy 5713 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2430 18.5 2 Light 552 4.2 3 Moderate 1040 7.9 4 Heavy 9087 69.3 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5760 (44%)
. Conso higher : 488 (3.7%)
. Sumalco higher: 6829 (52.2%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 320)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 393)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 790)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 246)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 512)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2066)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 99)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 28)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2748)
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
. conso: Heavy / sumalco: Non−drinker (n = 84)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 9)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 22)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2.1%
. Self−report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10342 56.9 3 Current HRT 6692 36.8 4 NA 2268 12.5


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 168 | Medium 168-284 | High >= 284 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1621 (33.3%) Medium 1615 (33.2%) High 1631 (33.5%)
WHO corrected (N = 4867)
  Low 1688 (34.7%) Medium 2052 (42.2%) High 1127 (23.2%)
. Visit: F2
Tertile (N = 4467)
  Low 1453 (32.5%) Medium 1558 (34.9%) High 1456 (32.6%)
WHO corrected (N = 4467)
  Low 1556 (34.8%) Medium 1935 (43.3%) High 976 (21.8%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                [31.5s, 3+, 19-]
. Visit F1: 4048/4867 in same category (83.2%)
. Visit F2: 3672/4467 in same category (82.2%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
```

```
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10247 50.1 2 >= 2 servings/day 2861 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11585 56.7 2 >= 2 servings/day 1523 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11770 (89.8%)
. Disagree: 1338 (10.2%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1338
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 487 9.8 2 F2 4582 505 11 3 F3 3575 346 9.7


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 109.5
. Q2 | Q3 boundary : 195.9
. Q3 | Q4 boundary : 301.7
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.1
. Q2 | Q3 boundary : 197.2
. Q3 | Q4 boundary : 293.8
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 845 17.3 12 F2 Q2 1366 27.9 13 F2 Q3
1272 26 14 F2 Q4 1077 22 15 F2 NA 334 6.8 16 F3 Q1 612 16.3 17 F3 Q2 1225 32.7
18 F3 Q3 1028 27.4 19 F3 Q4 777 20.7 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1452 28.7  7 F1 Q2 1043 20.6  8 F1 Q3 1141 22.5  9 F1
Q4 1315 26 10 F1 NA 113 2.2 11 F2 Q1 1046 21.4 12 F2 Q2 1187 24.3 13 F2 Q3 1188
24.3 14 F2 Q4 1139 23.3 15 F2 NA 334 6.8 16 F3 Q1 791 21.1 17 F3 Q2 1058 28.2
18 F3 Q3 959 25.6 19 F3 Q4 834 22.2 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
```

```
. Same quartile : 12310 (93.6%)
. Different quartile: 843 (6.4%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 594 2 Q3 Q2 53 3 Q3 Q4 196
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 310 6.3 2 F2 4560 285 6.2 3 F3 3642 248 6.8


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19504 2
derived 768 3 NA 170


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
```

```
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8770 | No: 11659 | Missing: 13
. Prevalence: 42.9% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8770
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[12/20] deriving ...
```

```
—> 45 targets ................                [31.5s, 3+, 19—]

.. Derive Education Level ..
—> 45 targets ................                [31.5s, 3+, 19—]


—> 45 targets ................                [31.5s, 3+, 19—]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


—> 45 targets ................                [31.5s, 3+, 19—]

.. Derive Alcohol Category ..
—> 45 targets ................                [31.5s, 3+, 19—]


—> 45 targets ................                [31.5s, 3+, 19—]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non—drinker
5545 27.6 2 Light 4374 21.7 3 Moderate 4490 22.3 4 Heavy 5710 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non—drinker 2456 18.7 2 Light 582 4.4 3 Moderate 1041 7.9 4 Heavy 9030 68.9 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5773 (44.1%)
. Conso higher : 525 (4%)
. Sumalco higher: 6779 (51.8%)
Breakdown of mismatches:

. conso: Non—drinker / sumalco: Light (n = 335)
—> 45 targets ................                [31.5s, 3+, 19—]

. conso: Non—drinker / sumalco: Moderate (n = 389)
—> 45 targets ................                [31.5s, 3+, 19—]

. conso: Non—drinker / sumalco: Heavy (n = 778)
—> 45 targets ................                [31.5s, 3+, 19—]

. conso: Light / sumalco: Non—drinker (n = 272)
—> 45 targets ................                [31.5s, 3+, 19—]

. conso: Light / sumalco: Moderate (n = 507)
—> 45 targets ................                [31.5s, 3+, 19—]

. conso: Light / sumalco: Heavy (n = 2026)
—> 45 targets ................                [31.5s, 3+, 19—]
```

```
. conso: Moderate / sumalco: Non−drinker (n = 110)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 30)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2744)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 80)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 9)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 24)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
```

```
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10265 56.5 3 Current HRT 6769 37.2 4 NA 2268 12.5


-> 45 targets ................                [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 168 | Medium 168-284 | High >= 284 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1609 (33.1%) Medium 1631 (33.5%) High 1627 (33.4%)
WHO corrected (N = 4867)
  Low 1679 (34.5%) Medium 2069 (42.5%) High 1119 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1453 (32.5%) Medium 1581 (35.4%) High 1433 (32.1%)
WHO corrected (N = 4467)
  Low 1545 (34.6%) Medium 1956 (43.8%) High 966 (21.6%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                [31.5s, 3+, 19-]
. Visit F1: 4041/4867 in same category (83.0%)
. Visit F2: 3710/4467 in same category (83.1%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
```

```
  non−fermented: 13077
  low−fat: 13080
  high−fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Cumulative average dairy derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_fermented_gday_cumavg: 13156
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_non_fermented_gday_cumavg: 13154
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_lowfat_gday_cumavg: 13156
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_highfat_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_total_nofondue_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Per−item dairy FFQ cumulative average derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ1amount_cumavg: 13170
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ2amount_cumavg: 13162
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ3amount_cumavg: 13167
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ4amount_cumavg: 13166
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ5amount_cumavg: 13163
−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
−> 45 targets ................                [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10258 50.2 2 >= 2 servings/day 2850 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11578 56.6 2 >= 2 servings/day 1530 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11788 (89.9%)
. Disagree: 1320 (10.1%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1320
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 485 9.8 2 F2 4582 483 10.5 3 F3 3575 352 9.8


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110
. Q2 | Q3 boundary : 195.3
. Q3 | Q4 boundary : 301.1
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.3
. Q2 | Q3 boundary : 197.6
. Q3 | Q4 boundary : 293.8
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 852 17.4 12 F2 Q2 1365 27.9 13 F2 Q3
1267 25.9 14 F2 Q4 1076 22 15 F2 NA 334 6.8 16 F3 Q1 609 16.2 17 F3 Q2 1177
31.4 18 F3 Q3 1066 28.4 19 F3 Q4 790 21.1 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
```

```
Baseline NA 6733 100  6 F1 Q1 1452 28.7  7 F1 Q2 1053 20.8  8 F1 Q3 1135 22.4  9 F1
Q4 1311 25.9 10 F1 NA 113 2.2 11 F2 Q1 1056 21.6 12 F2 Q2 1201 24.5 13 F2 Q3
1170 23.9 14 F2 Q4 1133 23.2 15 F2 NA 334 6.8 16 F3 Q1 781 20.8 17 F3 Q2 1034
27.6 18 F3 Q3 983 26.2 19 F3 Q4 844 22.5 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12281 (93.4%)
. Different quartile: 872 (6.6%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 590 2 Q3 Q2 98 3 Q3 Q4 184
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 316 6.4 2 F2 4560 301 6.6 3 F3 3642 255 7


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19497 2
derived 775 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²
```

```
-> 45 targets ...............                [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ...............                [31.5s, 3+, 19-]
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8786 | No: 11643 | Missing: 13
. Prevalence: 43% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8786
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[13/20] deriving ...


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5515 27.4 2 Light 4396 21.8 3 Moderate 4495 22.3 4 Heavy 5713 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2459 18.8 2 Light 569 4.3 3 Moderate 1038 7.9 4 Heavy 9043 69 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5748 (44%)
. Conso higher : 547 (4.2%)
. Sumalco higher: 6782 (51.9%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 324)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 390)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 769)
−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
. conso: Light / sumalco: Non−drinker (n = 282)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 510)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2044)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 115)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 37)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2745)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 79)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 9)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 25)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10269 56.5 3 Current HRT 6765 37.2 4 NA 2268 12.5


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 168 | Medium 168−283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1608 (33.0%) Medium 1636 (33.6%) High 1623 (33.3%)
WHO corrected (N = 4867)
  Low 1687 (34.7%) Medium 2061 (42.3%) High 1119 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1442 (32.3%) Medium 1610 (36.0%) High 1415 (31.7%)
WHO corrected (N = 4467)
  Low 1533 (34.3%) Medium 1946 (43.6%) High 988 (22.1%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Method alignment (tertile vs WHO corrected)
−> 45 targets ................                 [31.5s, 3+, 19−]
. Visit F1: 4034/4867 in same category (82.9%)
. Visit F2: 3749/4467 in same category (83.9%)


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Dairy sub−category derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non−fermented: 13077
  low−fat: 13080
  high−fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Cumulative average dairy derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_fermented_gday_cumavg: 13156
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_non_fermented_gday_cumavg: 13154
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_lowfat_gday_cumavg: 13156
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_highfat_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]

  dairy_total_nofondue_gday_cumavg: 13153
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Per−item dairy FFQ cumulative average derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ1amount_cumavg: 13170
−> 45 targets ................                 [31.5s, 3+, 19−]

  FFQ2amount_cumavg: 13162
−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
  FFQ3amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
```

```
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10255 50.2 2 >= 2 servings/day 2853 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11587 56.7 2 >= 2 servings/day 1521 7.4 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
. Agree : 11776 (89.8%)
. Disagree: 1332 (10.2%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1332
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 485 9.8 2 F2 4582 484 10.6 3 F3 3575 363 10.2


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Quartiles ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Baseline quartile cut-points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.3
. Q2 | Q3 boundary : 196.8
. Q3 | Q4 boundary : 302.4
. Overall quartile cut-points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.1
. Q2 | Q3 boundary : 198.9
. Q3 | Q4 boundary : 295.4
. Distribution of ‘dairy_quartile_baseline‘ by visit:
```

```
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 873 17.8 12 F2 Q2 1327 27.1 13 F2 Q3
1272 26 14 F2 Q4 1088 22.2 15 F2 NA 334 6.8 16 F3 Q1 632 16.8 17 F3 Q2 1165
31.1 18 F3 Q3 1043 27.8 19 F3 Q4 802 21.4 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1443 28.5  7 F1 Q2 1065 21  8 F1 Q3 1141 22.5  9 F1
Q4 1302 25.7 10 F1 NA 113 2.2 11 F2 Q1 1057 21.6 12 F2 Q2 1180 24.1 13 F2 Q3
1177 24 14 F2 Q4 1146 23.4 15 F2 NA 334 6.8 16 F3 Q1 789 21 17 F3 Q2 1043 27.8
18 F3 Q3 970 25.9 19 F3 Q4 840 22.4 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12343 (93.8%)
. Different quartile: 810 (6.2%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 546 2 Q3 Q2 104 3 Q3 Q4 160
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 301 6.1 2 F2 4560 279 6.1 3 F3 3642 230 6.3


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19498 2
derived 774 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
```

```
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.8%


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Age ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ................                [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
```

```
  alcohol : 10810


-> 45 targets ...............                 [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ...............                 [31.5s, 3+, 19-]


-> 45 targets ...............                 [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8770 | No: 11659 | Missing: 13
. Prevalence: 42.9% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8770
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[14/20] deriving ...


-> 45 targets ...............                 [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ...............                 [31.5s, 3+, 19-]


-> 45 targets ...............                 [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ...............                 [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ...............                 [31.5s, 3+, 19-]


-> 45 targets ...............                 [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5553 27.6 2 Light 4408 21.9 3 Moderate 4461 22.2 4 Heavy 5697 28.3 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2430 18.5 2 Light 583 4.4 3 Moderate 1044 8 4 Heavy 9052 69.1 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5735 (43.9%)
. Conso higher : 509 (3.9%)
. Sumalco higher: 6833 (52.3%)
Breakdown of mismatches:
```

```
. conso: Non−drinker / sumalco: Light (n = 335)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 394)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 793)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 260)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 524)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2053)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 109)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 30)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2734)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 79)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 8)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 23)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2.1%
. Self−report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                [31.5s, 3+, 19−]
```

```
.. Derive CVD Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10250 56.4 3 Current HRT 6784 37.3 4 NA 2268 12.5


-> 45 targets ................                [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 169 | Medium 169-284 | High >= 284 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1620 (33.3%) Medium 1623 (33.3%) High 1624 (33.4%)
WHO corrected (N = 4867)
  Low 1671 (34.3%) Medium 2078 (42.7%) High 1118 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1505 (33.7%) Medium 1547 (34.6%) High 1415 (31.7%)
WHO corrected (N = 4467)
  Low 1586 (35.5%) Medium 1904 (42.6%) High 977 (21.9%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                [31.5s, 3+, 19-]
. Visit F1: 4052/4867 in same category (83.3%)
. Visit F2: 3726/4467 in same category (83.4%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Dairy protein content derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Non−dairy protein cumulative average derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10261 50.2 2 >= 2 servings/day 2847 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11576 56.6 2 >= 2 servings/day 1532 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11793 (90%)
. Disagree: 1315 (10%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1315
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 491 9.9 2 F2 4582 484 10.6 3 F3 3575 340 9.5


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.1
. Q2 | Q3 boundary : 196.1
. Q3 | Q4 boundary : 302.2
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.5
. Q2 | Q3 boundary : 198.5
. Q3 | Q4 boundary : 294.8
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 870 17.8 12 F2 Q2 1326 27.1 13 F2 Q3
1277 26.1 14 F2 Q4 1087 22.2 15 F2 NA 334 6.8 16 F3 Q1 641 17.1 17 F3 Q2 1152
30.7 18 F3 Q3 1063 28.3 19 F3 Q4 786 21 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1434 28.3  7 F1 Q2 1078 21.3  8 F1 Q3 1128 22.3  9 F1
Q4 1311 25.9 10 F1 NA 113 2.2 11 F2 Q1 1063 21.7 12 F2 Q2 1170 23.9 13 F2 Q3
1178 24.1 14 F2 Q4 1149 23.5 15 F2 NA 334 6.8 16 F3 Q1 792 21.1 17 F3 Q2 1040
27.7 18 F3 Q3 982 26.2 19 F3 Q4 828 22.1 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12324 (93.7%)
. Different quartile: 829 (6.3%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 540 2 Q3 Q2 112 3 Q3 Q4 177
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 305 6.2 2 F2 4560 292 6.4 3 F3 3642 232 6.4


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19506 2
derived 766 3 NA 170
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.3 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
-> 45 targets ...............                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8780 | No: 11649 | Missing: 13
. Prevalence: 43% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8780
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[15/20] deriving ...


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ...............                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ...............                [31.5s, 3+, 19-]


-> 45 targets ...............                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5532 27.5 2 Light 4388 21.8 3 Moderate 4474 22.2 4 Heavy 5725 28.5 5 NA 323 1.6
```

```
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2451 18.7 2 Light 567 4.3 3 Moderate 1020 7.8 4 Heavy 9071 69.2 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5773 (44.1%)
. Conso higher : 504 (3.9%)
. Sumalco higher: 6800 (52%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 334)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 375)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 770)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 262)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 514)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2052)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 100)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 26)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2755)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 85)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 8)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 23)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
```

```
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2.1%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ...............                [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ...............                [31.5s, 3+, 19−]


−> 45 targets ...............                [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2987 rows | 1031 / 6746 participants
. No : 17439 rows


−> 45 targets ...............                [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ...............                [31.5s, 3+, 19−]


−> 45 targets ...............                [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10267 56.5 3 Current HRT 6767 37.2 4 NA 2268 12.5


−> 45 targets ...............                [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ...............                [31.5s, 3+, 19−]


−> 45 targets ...............                [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 168 | Medium 168−285 | High >= 285 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1608 (33.0%) Medium 1634 (33.6%) High 1625 (33.4%)
WHO corrected (N = 4867)
  Low 1682 (34.6%) Medium 2082 (42.8%) High 1103 (22.7%)
. Visit: F2
Tertile (N = 4467)
  Low 1481 (33.2%) Medium 1563 (35.0%) High 1423 (31.9%)
```

```
WHO corrected (N = 4467)
  Low 1569 (35.1%) Medium 1923 (43.0%) High 975 (21.8%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4035/4867 in same category (82.9%)
. Visit F2: 3709/4467 in same category (83.0%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10267 50.2 2 >= 2 servings/day 2841 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11605 56.8 2 >= 2 servings/day 1503 7.4 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
. Agree : 11770 (89.8%)
. Disagree: 1338 (10.2%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1338
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 497 10 2 F2 4582 487 10.6 3 F3 3575 354 9.9
```

```
−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.4
. Q2 | Q3 boundary : 196.3
. Q3 | Q4 boundary : 301.8
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.4
. Q2 | Q3 boundary : 198.4
. Q3 | Q4 boundary : 294.5
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 865 17.7 12 F2 Q2 1334 27.3 13 F2 Q3
1271 26 14 F2 Q4 1090 22.3 15 F2 NA 334 6.8 16 F3 Q1 626 16.7 17 F3 Q2 1182
31.5 18 F3 Q3 1049 28 19 F3 Q4 785 20.9 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1438 28.4  7 F1 Q2 1064 21  8 F1 Q3 1138 22.5  9 F1
Q4 1311 25.9 10 F1 NA 113 2.2 11 F2 Q1 1061 21.7 12 F2 Q2 1166 23.8 13 F2 Q3
1190 24.3 14 F2 Q4 1143 23.4 15 F2 NA 334 6.8 16 F3 Q1 790 21.1 17 F3 Q2 1058
28.2 18 F3 Q3 960 25.6 19 F3 Q4 834 22.2 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12324 (93.7%)
. Different quartile: 829 (6.3%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 560 2 Q3 Q2 94 3 Q3 Q4 175
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 299 6 2 F2 4560 277 6.1 3 F3 3642 253 6.9


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Derive Smoking Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19474 2
derived 798 3 NA 170


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
```

```
. Max absolute difference : 5 years


-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8777 | No: 11652 | Missing: 13
. Prevalence: 43% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8777
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[16/20] deriving ...


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Education Level ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5537 27.5 2 Light 4366 21.7 3 Moderate 4498 22.4 4 Heavy 5718 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2412 18.4 2 Light 568 4.3 3 Moderate 1047 8 4 Heavy 9082 69.3 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5733 (43.8%)
. Conso higher : 507 (3.9%)
. Sumalco higher: 6837 (52.3%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 322)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 406)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 794)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 255)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 510)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2035)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 101)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 31)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2770)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 90)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 10)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 20)
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
.. Derive Diabetes Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2%
. Self-report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived 'cdv_event' and
'cvdbase_adj'.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10319 56.8 3 Current HRT 6715 36.9 4 NA 2268 12.5


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 168 | Medium 168-283 | High >= 283 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
```

```
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1611 (33.1%) Medium 1625 (33.4%) High 1631 (33.5%)
WHO corrected (N = 4867)
  Low 1672 (34.4%) Medium 2082 (42.8%) High 1113 (22.9%)
. Visit: F2
Tertile (N = 4467)
  Low 1458 (32.6%) Medium 1591 (35.6%) High 1418 (31.7%)
WHO corrected (N = 4467)
  Low 1543 (34.5%) Medium 1964 (44.0%) High 960 (21.5%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4052/4867 in same category (83.3%)
. Visit F2: 3712/4467 in same category (83.1%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10252 50.2 2 >= 2 servings/day 2856 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11603 56.8 2 >= 2 servings/day 1505 7.4 3 NA 7334 35.9
```

```
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11757 (89.7%)
. Disagree: 1351 (10.3%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1351
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 490 9.9 2 F2 4582 513 11.2 3 F3 3575 348 9.7


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.7
. Q2 | Q3 boundary : 196.3
. Q3 | Q4 boundary : 303.2
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.1
. Q2 | Q3 boundary : 199.1
. Q3 | Q4 boundary : 296.1
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 876 17.9 12 F2 Q2 1305 26.7 13 F2 Q3
1281 26.2 14 F2 Q4 1098 22.4 15 F2 NA 334 6.8 16 F3 Q1 625 16.7 17 F3 Q2 1165
31.1 18 F3 Q3 1059 28.2 19 F3 Q4 793 21.1 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1435 28.3  7 F1 Q2 1080 21.3  8 F1 Q3 1137 22.5  9 F1
Q4 1299 25.7 10 F1 NA 113 2.2 11 F2 Q1 1057 21.6 12 F2 Q2 1162 23.7 13 F2 Q3
1190 24.3 14 F2 Q4 1151 23.5 15 F2 NA 334 6.8 16 F3 Q1 797 21.2 17 F3 Q2 1046
27.9 18 F3 Q3 961 25.6 19 F3 Q4 838 22.3 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12314 (93.6%)
. Different quartile: 839 (6.4%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 550 2 Q3 Q2 130 3 Q3 Q4 159
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 297 6 2 F2 4560 272 6 3 F3 3642 270 7.4


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive ATCs ..
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_atc: ATC-based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Smoking Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19492 2
derived 780 3 NA 170


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 12 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.8%


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ................                [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8763 | No: 11666 | Missing: 13
. Prevalence: 42.9% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8763
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[17/20] deriving ...


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
```

```
  everywhere).

-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5545 27.6 2 Light 4374 21.7 3 Moderate 4460 22.2 4 Heavy 5740 28.5 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2449 18.7 2 Light 563 4.3 3 Moderate 1018 7.8 4 Heavy 9079 69.3 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5778 (44.2%)
. Conso higher : 516 (3.9%)
. Sumalco higher: 6783 (51.9%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 330)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 382)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 794)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 274)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 503)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2039)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 101)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 27)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2735)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 84)
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
. conso: Heavy / sumalco: Light (n = 9)
−> 45 targets ................                 [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 21)
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2%
. Self−report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive HRT Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10278 56.6 3 Current HRT 6756 37.2 4 NA 2268 12.5


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 170 | Medium 170-286 | High >= 286 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1617 (33.2%) Medium 1623 (33.3%) High 1627 (33.4%)
WHO corrected (N = 4867)
  Low 1651 (33.9%) Medium 2086 (42.9%) High 1130 (23.2%)
. Visit: F2
Tertile (N = 4467)
  Low 1492 (33.4%) Medium 1567 (35.1%) High 1408 (31.5%)
WHO corrected (N = 4467)
  Low 1544 (34.6%) Medium 1929 (43.2%) High 994 (22.3%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
. Visit F1: 4074/4867 in same category (83.7%)
. Visit F2: 3755/4467 in same category (84.1%)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10266 50.2 2 >= 2 servings/day 2842 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11574 56.6 2 >= 2 servings/day 1534 7.5 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11800 (90%)
. Disagree: 1308 (10%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1308
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 496 10 2 F2 4582 482 10.5 3 F3 3575 330 9.2


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 109.8
. Q2 | Q3 boundary : 195.9
. Q3 | Q4 boundary : 302.4
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123.3
. Q2 | Q3 boundary : 198.7
. Q3 | Q4 boundary : 295.2
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 847 17.3 12 F2 Q2 1340 27.4 13 F2 Q3
1285 26.3 14 F2 Q4 1088 22.2 15 F2 NA 334 6.8 16 F3 Q1 618 16.5 17 F3 Q2 1184
31.6 18 F3 Q3 1039 27.7 19 F3 Q4 801 21.4 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1447 28.6  7 F1 Q2 1070 21.1  8 F1 Q3 1132 22.4  9 F1
Q4 1302 25.7 10 F1 NA 113 2.2 11 F2 Q1 1056 21.6 12 F2 Q2 1169 23.9 13 F2 Q3
1187 24.3 14 F2 Q4 1148 23.5 15 F2 NA 334 6.8 16 F3 Q1 786 21 17 F3 Q2 1049 28
18 F3 Q3 969 25.8 19 F3 Q4 838 22.3 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12294 (93.5%)
. Different quartile: 859 (6.5%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
```

```
<int> 1 Q2 Q1 586 2 Q3 Q2 112 3 Q3 Q4 161
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 314 6.3 2 F2 4560 307 6.7 3 F3 3642 238 6.5


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive ATCs ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_atc: ATC-based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Smoking Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19507 2
derived 765 3 NA 170


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive BMI ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.6 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.9%
```

```
-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8781 | No: 11648 | Missing: 13
. Prevalence: 43% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8781
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
[18/20] deriving ...


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
.. Derive Education Level ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Alcohol Category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non-drinker
5560 27.6 2 Light 4373 21.7 3 Moderate 4471 22.2 4 Heavy 5715 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non-drinker 2424 18.5 2 Light 550 4.2 3 Moderate 1055 8 4 Heavy 9080 69.3 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5744 (43.9%)
. Conso higher : 503 (3.8%)
. Sumalco higher: 6830 (52.2%)
Breakdown of mismatches:

. conso: Non-drinker / sumalco: Light (n = 321)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Moderate (n = 395)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Non-drinker / sumalco: Heavy (n = 822)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Non-drinker (n = 258)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Moderate (n = 521)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2036)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 109)
-> 45 targets ................                [31.5s, 3+, 19-]
```

```
. conso: Moderate / sumalco: Light (n = 23)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2735)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 84)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 9)
-> 45 targets ................                [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 20)
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Diabetes Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2.1%
. Self-report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
```

```
6.3 2 Past HRT 10263 56.5 3 Current HRT 6771 37.3 4 NA 2268 12.5


−> 45 targets ................                [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 168 | Medium 168−282 | High >= 282 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1620 (33.3%) Medium 1619 (33.3%) High 1628 (33.4%)
WHO corrected (N = 4867)
  Low 1684 (34.6%) Medium 2066 (42.4%) High 1117 (23.0%)
. Visit: F2
Tertile (N = 4467)
  Low 1506 (33.7%) Medium 1486 (33.3%) High 1475 (33.0%)
WHO corrected (N = 4467)
  Low 1575 (35.3%) Medium 1887 (42.2%) High 1005 (22.5%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


−> 45 targets ................                [31.5s, 3+, 19−]

.. Method alignment (tertile vs WHO corrected)
−> 45 targets ................                [31.5s, 3+, 19−]
. Visit F1: 4050/4867 in same category (83.2%)
. Visit F2: 3704/4467 in same category (82.9%)


−> 45 targets ................                [31.5s, 3+, 19−]

.. Dairy sub−category derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non−fermented: 13077
  low−fat: 13080
  high−fat: 13075
  total (excl. fondue): 13076
```

```
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151
```

```
−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10274 50.3 2 >= 2 servings/day 2834 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11610 56.8 2 >= 2 servings/day 1498 7.3 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11772 (89.8%)
. Disagree: 1336 (10.2%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1336
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 495 10 2 F2 4582 490 10.7 3 F3 3575 351 9.8


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 109.8
. Q2 | Q3 boundary : 195.3
. Q3 | Q4 boundary : 302.6
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.7
. Q2 | Q3 boundary : 198.2
. Q3 | Q4 boundary : 294.7
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 867 17.7 12 F2 Q2 1317 26.9 13 F2 Q3
1307 26.7 14 F2 Q4 1069 21.8 15 F2 NA 334 6.8 16 F3 Q1 611 16.3 17 F3 Q2 1193
31.8 18 F3 Q3 1050 28 19 F3 Q4 788 21 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1445 28.5  7 F1 Q2 1066 21.1  8 F1 Q3 1123 22.2  9 F1
Q4 1317 26 10 F1 NA 113 2.2 11 F2 Q1 1066 21.8 12 F2 Q2 1163 23.8 13 F2 Q3 1198
24.5 14 F2 Q4 1133 23.2 15 F2 NA 334 6.8 16 F3 Q1 778 20.7 17 F3 Q2 1059 28.2
18 F3 Q3 967 25.8 19 F3 Q4 838 22.3 20 F3 NA 109 2.9
```

```
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12274 (93.3%)
. Different quartile: 879 (6.7%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 573 2 Q3 Q2 113 3 Q3 Q4 193
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 321 6.5 2 F2 4560 308 6.8 3 F3 3642 250 6.9


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19507 2
derived 765 3 NA 170


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.1 − 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Deriving BMI category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.8%


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Age ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ................                [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive htn Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8773 | No: 11656 | Missing: 13
. Prevalence: 42.9% (among non−missing rows)
```

```
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8773
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[19/20] deriving ...


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5534 27.5 2 Light 4393 21.8 3 Moderate 4447 22.1 4 Heavy 5745 28.6 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2446 18.7 2 Light 562 4.3 3 Moderate 1030 7.9 4 Heavy 9071 69.2 5
NA 7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5787 (44.3%)
. Conso higher : 519 (4%)
. Sumalco higher: 6771 (51.8%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 317)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Moderate (n = 389)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 792)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 271)
−> 45 targets ................                [31.5s, 3+, 19−]
```

```
. conso: Light / sumalco: Moderate (n = 508)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Light / sumalco: Heavy (n = 2050)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Non-drinker (n = 103)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Light (n = 29)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Moderate / sumalco: Heavy (n = 2715)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Non-drinker (n = 85)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Light (n = 13)
-> 45 targets ................                 [31.5s, 3+, 19-]

. conso: Heavy / sumalco: Moderate (n = 18)
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive Diabetes Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.4%
. Disagreement between FPG and HbA1c: 2%
. Self-report vs objective disagreement: 4.4%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive CVD Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
.. Derive HRT Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10220 56.2 3 Current HRT 6814 37.5 4 NA 2268 12.5


−> 45 targets ................                [31.5s, 3+, 19−]

.. Physical Activity Derivation Report ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Tertile cut−points (from F1, N = 4867):
  Low < 168 | Medium 168−285 | High >= 285 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192−383 | High > 383 ME−min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1613 (33.1%) Medium 1631 (33.5%) High 1623 (33.3%)
WHO corrected (N = 4867)
  Low 1684 (34.6%) Medium 2055 (42.2%) High 1128 (23.2%)
. Visit: F2
Tertile (N = 4467)
  Low 1516 (33.9%) Medium 1585 (35.5%) High 1366 (30.6%)
WHO corrected (N = 4467)
  Low 1597 (35.8%) Medium 1944 (43.5%) High 926 (20.7%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


−> 45 targets ................                [31.5s, 3+, 19−]

.. Method alignment (tertile vs WHO corrected)
−> 45 targets ................                [31.5s, 3+, 19−]
. Visit F1: 4061/4867 in same category (83.4%)
. Visit F2: 3720/4467 in same category (83.3%)


−> 45 targets ................                [31.5s, 3+, 19−]

.. Dairy sub−category derivation ..
−> 45 targets ................                [31.5s, 3+, 19−]
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ1amount_cumavg: 13170
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
  FFQ4amount_cumavg: 13166
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                 [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
.. Non−dairy protein cumulative average derivation ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Servings ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10269 50.2 2 >= 2 servings/day 2839 13.9 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11608 56.8 2 >= 2 servings/day 1500 7.3 3 NA 7334 35.9
. Method agreement (rows with both non−missing, n = 13108):
. Agree : 11769 (89.8%)
. Disagree: 1339 (10.2%)
Cross−tabulation of disagreements (freq −> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1339
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 497 10 2 F2 4582 493 10.8 3 F3 3575 349 9.8


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Dairy Quartiles ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Baseline quartile cut−points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.2
. Q2 | Q3 boundary : 196.4
. Q3 | Q4 boundary : 303.3
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 122.9
. Q2 | Q3 boundary : 198.8
. Q3 | Q4 boundary : 296.1
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 871 17.8 12 F2 Q2 1327 27.1 13 F2 Q3
```

```
1269 25.9 14 F2 Q4 1093 22.3 15 F2 NA 334 6.8 16 F3 Q1 626 16.7 17 F3 Q2 1165
31.1 18 F3 Q3 1059 28.2 19 F3 Q4 792 21.1 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1434 28.3  7 F1 Q2 1075 21.2  8 F1 Q3 1140 22.5  9 F1
Q4 1302 25.7 10 F1 NA 113 2.2 11 F2 Q1 1069 21.8 12 F2 Q2 1167 23.8 13 F2 Q3
1178 24.1 14 F2 Q4 1146 23.4 15 F2 NA 334 6.8 16 F3 Q1 786 21 17 F3 Q2 1046
27.9 18 F3 Q3 970 25.9 19 F3 Q4 840 22.4 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12322 (93.7%)
. Different quartile: 831 (6.3%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 554 2 Q3 Q2 112 3 Q3 Q4 165
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 293 5.9 2 F2 4560 289 6.3 3 F3 3642 249 6.8


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19505 2
derived 767 3 NA 170


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive BMI ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non−missing): 13.7 − 79.9 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non−missing, n = 19759):
```

```
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01−0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Deriving BMI category ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.6%
  Prevalence of Overweight + Obesity: 55.7%


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Age ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non−missing): 34.9 − 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non−missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1−1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Food group derivation (updated MSM structure) ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
.. Derive htn Status ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. htn_status derived.
. Total rows: 20442 | Yes: 8776 | No: 11653 | Missing: 13
. Prevalence: 43% (among non−missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8776
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self−reported, tier 3) : 0
[20/20] deriving ...


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Education Level ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. derive_education: education_level derived and propagated.
. Participants with education_level: 6722 / 6746 (99.6%).
. Rows with education_level filled (all visits): 20415.
! 24 participant(s) have no education_level at any visit (edtyp4 missing
  everywhere).


−> 45 targets ................                 [31.5s, 3+, 19−]

.. Derive Alcohol Category ..
−> 45 targets ................                 [31.5s, 3+, 19−]


−> 45 targets ................                 [31.5s, 3+, 19−]
. Total rows: 20442
. conso_hebdo present (primary source): 20119 (98.4%)
. sumalco present (comparison source) : 13109 (64.1%)
. alcohol_category_conso derived : 20119 (98.4%)
. alcohol_category_sumalco derived: 13109 (64.1%)
. Distribution of alcohol_category_conso:
# A tibble: 5 × 3 alcohol_category_conso n pct <ord> <int> <dbl> 1 Non−drinker
5538 27.5 2 Light 4404 21.9 3 Moderate 4462 22.2 4 Heavy 5715 28.4 5 NA 323 1.6
. Distribution of alcohol_category_sumalco:
# A tibble: 5 × 3 alcohol_category_sumalco n pct <ord> <int> <dbl> 1
Non−drinker 2474 18.9 2 Light 570 4.3 3 Moderate 1020 7.8 4 Heavy 9045 69 5 NA
7333 55.9
. Source agreement (rows where both present, n = 13077):
. Agree : 5780 (44.2%)
. Conso higher : 518 (4%)
. Sumalco higher: 6779 (51.8%)
Breakdown of mismatches:

. conso: Non−drinker / sumalco: Light (n = 325)
−> 45 targets ................                 [31.5s, 3+, 19−]
```

```
. conso: Non−drinker / sumalco: Moderate (n = 381)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Non−drinker / sumalco: Heavy (n = 770)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Non−drinker (n = 275)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Moderate (n = 515)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Light / sumalco: Heavy (n = 2045)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Non−drinker (n = 101)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Light (n = 29)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Moderate / sumalco: Heavy (n = 2743)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Non−drinker (n = 85)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Light (n = 8)
−> 45 targets ................                [31.5s, 3+, 19−]

. conso: Heavy / sumalco: Moderate (n = 20)
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Diabetes Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. Diabetes derivation completed
. Total measurements: 20442
. No data on any source: 28 (0.1%)
. Sources available (1/2/3): 255 / 11739 / 8420
. Disagreement between any sources: 4.3%
. Disagreement between FPG and HbA1c: 2.1%
. Self−report vs objective disagreement: 4.3%
derive_cvd: 23 Baseline row(s) mismatch between derived ‘cdv_event‘ and
‘cvdbase_adj‘.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive CVD Status ..
−> 45 targets ................                [31.5s, 3+, 19−]
```

```
-> 45 targets ................                 [31.5s, 3+, 19-]
. cdv_event derived from 13 flag(s) and carried forward.
. Total rows: 20442 | derived: 20426 | missing: 16
. Yes (any CVD event): 2983 rows | 1030 / 6746 participants
. No : 17443 rows


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive HRT Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. hrt_status derived.
. Total rows: 20442 | derived: 18174 | missing (esthrp NA): 2268
. Distribution (among derived rows):
# A tibble: 4 × 3 hrt_status n pct <fct> <int> <dbl> 1 Never / Not current 1140
6.3 2 Past HRT 10290 56.6 3 Current HRT 6744 37.1 4 NA 2268 12.5


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Physical Activity Derivation Report ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. Tertile cut-points (from F1, N = 4867):
  Low < 169 | Medium 169-284 | High >= 284 min/day MVPA
. WHO thresholds (x1.28 correction, collapsed to 3 levels):
  Low <= 191 | Medium 192-383 | High > 383 ME-min/week
. Visit: Baseline
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
. Visit: F1
Tertile (N = 4867)
  Low 1607 (33.0%) Medium 1628 (33.4%) High 1632 (33.5%)
WHO corrected (N = 4867)
  Low 1679 (34.5%) Medium 2072 (42.6%) High 1116 (22.9%)
. Visit: F2
Tertile (N = 4467)
  Low 1479 (33.1%) Medium 1610 (36.0%) High 1378 (30.8%)
WHO corrected (N = 4467)
  Low 1576 (35.3%) Medium 1953 (43.7%) High 938 (21.0%)
. Visit: F3
Tertile (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)
WHO corrected (N = 0)
  Low 0 (NA) Medium 0 (NA) High 0 (NA)


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Method alignment (tertile vs WHO corrected)
-> 45 targets ................                 [31.5s, 3+, 19-]
```

```
. Visit F1: 4029/4867 in same category (82.8%)
. Visit F2: 3718/4467 in same category (83.2%)


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy sub-category derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy: derived variables added.
  total: 13075
  fermented: 13079
  non-fermented: 13077
  low-fat: 13080
  high-fat: 13075
  total (excl. fondue): 13076
! .visit is an unordered factor. Using current factor level order.


-> 45 targets ................                [31.5s, 3+, 19-]

.. Cumulative average dairy derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_cumavg: cumulative averages added.
. Visit order: Baseline < F1 < F2 < F3

  dairy_total_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_fermented_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_non_fermented_gday_cumavg: 13154
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_lowfat_gday_cumavg: 13156
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_highfat_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]

  dairy_total_nofondue_gday_cumavg: 13153
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Per-item dairy FFQ cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
```

```
  FFQ1amount_cumavg: 13170
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ2amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ3amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ4amount_cumavg: 13166
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ5amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ6amount_cumavg: 13167
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ7amount_cumavg: 13174
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ8amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ52amount_cumavg: 13169
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ53amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ63amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ68amount_cumavg: 13164
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ71amount_cumavg: 13161
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ82amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ83amount_cumavg: 13159
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ84amount_cumavg: 13162
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ85amount_cumavg: 13160
-> 45 targets ................                [31.5s, 3+, 19-]

  FFQ86amount_cumavg: 13163
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]

.. Dairy protein content derivation ..
```

```
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_dairy_protein: protein content columns added.
. prot_content_dairy_cumavg valid: 13153


-> 45 targets ................                [31.5s, 3+, 19-]

.. Non-dairy protein cumulative average derivation ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_nondairy_protein_cumavg: columns added.
. sumprot1_cumavg valid: 13185
. prot_content_nondairy_cumavg valid: 13151


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Servings ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Total rows: 20442
. dairy_freq_total derived : 13108 (64.1%)
. dairy_portion_total derived : 13108 (64.1%)
. Distribution of ‘dairy_guidelines_freq‘:
# A tibble: 3 × 3 dairy_guidelines_freq n pct <fct> <int> <dbl> 1 < 2
servings/day 10251 50.1 2 >= 2 servings/day 2857 14 3 NA 7334 35.9
. Distribution of ‘dairy_guidelines_port‘:
# A tibble: 3 × 3 dairy_guidelines_port n pct <fct> <int> <dbl> 1 < 2
servings/day 11596 56.7 2 >= 2 servings/day 1512 7.4 3 NA 7334 35.9
. Method agreement (rows with both non-missing, n = 13108):
. Agree : 11763 (89.7%)
. Disagree: 1345 (10.3%)
Cross-tabulation of disagreements (freq -> portion):
# A tibble: 1 × 3 dairy_guidelines_freq dairy_guidelines_port n <fct> <fct>
<int> 1 >= 2 servings/day < 2 servings/day 1345
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 488 9.9 2 F2 4582 499 10.9 3 F3 3575 358 10


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Dairy Quartiles ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Baseline quartile cut-points (g/day, n = 4951):
. Q1 | Q2 boundary : 110.3
. Q2 | Q3 boundary : 196.1
. Q3 | Q4 boundary : 301.2
```

```
. Overall quartile cut−points (g/day, n = 13153):
. Q1 | Q2 boundary : 123
. Q2 | Q3 boundary : 198.5
. Q3 | Q4 boundary : 294.8
. Distribution of ‘dairy_quartile_baseline‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_baseline n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1238 24.4  7 F1 Q2 1238 24.4  8 F1 Q3 1237 24.4  9 F1
Q4 1238 24.4 10 F1 NA 113 2.2 11 F2 Q1 863 17.6 12 F2 Q2 1328 27.1 13 F2 Q3
1290 26.4 14 F2 Q4 1079 22 15 F2 NA 334 6.8 16 F3 Q1 633 16.9 17 F3 Q2 1172
31.2 18 F3 Q3 1031 27.5 19 F3 Q4 806 21.5 20 F3 NA 109 2.9
. Distribution of ‘dairy_quartile_overall‘ by visit:
# A tibble: 20 × 4 .visit dairy_quartile_overall n pct <ord> <ord> <int> <dbl>
1 Baseline Q1 0 0  2 Baseline Q2 0 0  3 Baseline Q3 0 0  4 Baseline Q4 0 0  5
Baseline NA 6733 100  6 F1 Q1 1433 28.3  7 F1 Q2 1074 21.2  8 F1 Q3 1138 22.5  9 F1
Q4 1306 25.8 10 F1 NA 113 2.2 11 F2 Q1 1061 21.7 12 F2 Q2 1171 23.9 13 F2 Q3
1199 24.5 14 F2 Q4 1129 23.1 15 F2 NA 334 6.8 16 F3 Q1 795 21.2 17 F3 Q2 1043
27.8 18 F3 Q3 951 25.4 19 F3 Q4 853 22.7 20 F3 NA 109 2.9
. Dairy quartiles derived.
. dairy_quartile_baseline: 13153 / 20442 rows non−missing
. dairy_quartile_overall : 13153 / 20442 rows non−missing
. Baseline vs overall quartile agreement (rows with both non−missing, n =
  13153):
. Same quartile : 12328 (93.7%)
. Different quartile: 825 (6.3%)
Cross−tabulation of differing classifications (baseline −> overall):
# A tibble: 3 × 3 dairy_quartile_baseline dairy_quartile_overall n <ord> <ord>
<int> 1 Q2 Q1 555 2 Q3 Q2 105 3 Q3 Q4 165
Disagreement rate by visit:
# A tibble: 3 × 4 .visit n n_differ pct_differ <ord> <int> <int> <dbl> 1 F1
4951 294 5.9 2 F2 4560 289 6.3 3 F3 3642 242 6.6


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive ATCs ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
. derive_atc: ATC−based flags derived.
. Processed 20442 rows; 14079 rows had valid ATC mapping data.


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive Smoking Status ..
−> 45 targets ................                [31.5s, 3+, 19−]


−> 45 targets ................                [31.5s, 3+, 19−]
# A tibble: 3 × 2 smoking_impute_source n <fct> <int> 1 observed 19490 2
derived 782 3 NA 170


−> 45 targets ................                [31.5s, 3+, 19−]

.. Derive BMI ..
```

```
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. BMI derived from Weight and Height.
. Total rows: 20442 | derived: 20176 | missing: 266
. BMI range (non-missing): 13.7 - 79.7 kg/m²
. Mean ± SD: 26.2 ± 4.7 kg/m²
. Agreement with BMI_source (rows with both non-missing, n = 19759):
. Difference < 0.01 (effectively identical): 19759 (100%)
. Difference 0.01-0.5 (minor rounding): 0 (0%)
. Difference >= 0.5 (meaningful mismatch): 0 (0%)
. Median absolute difference: 0.003 kg/m²
. Max absolute difference : 0.005 kg/m²


-> 45 targets ................                [31.5s, 3+, 19-]

.. Deriving BMI category ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. derive_bmi_category: BMI categories derived.
. Processed 20442 rows (266 missing BMI).
  Prevalence of Obesity: 17.5%
  Prevalence of Overweight + Obesity: 55.9%


-> 45 targets ................                [31.5s, 3+, 19-]

.. Derive Age ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Age derived from datbirth (Baseline, filled forward/back) and exam_date_iso.
. Total rows: 20442 | derived: 20177 | missing: 265
. Age range (non-missing): 34.9 - 90.1 years
. Mean ± SD: 58.6 ± 11.6 years
. Agreement with Age_source (rows with both non-missing, n = 20177):
. Difference < 0.1 yr (effectively identical): 19638 (97.3%)
. Difference 0.1-1 yr (minor rounding/timing): 536 (2.7%)
. Difference >= 1 yr (meaningful mismatch) : 3 (0%)
. Median absolute difference: 0 years
. Max absolute difference : 5 years


-> 45 targets ................                [31.5s, 3+, 19-]

.. Food group derivation (updated MSM structure) ..
-> 45 targets ................                [31.5s, 3+, 19-]


-> 45 targets ................                [31.5s, 3+, 19-]
. Derived food groups successfully.
  animal protein : 10730
  plant protein : 10825
```

```
  vegetables : 10748
  fruits : 10768
  grains : 10752
  fats : 10728
  processed : 10736
  alcohol : 10810


-> 45 targets ................                 [31.5s, 3+, 19-]

.. Derive htn Status ..
-> 45 targets ................                 [31.5s, 3+, 19-]


-> 45 targets ................                 [31.5s, 3+, 19-]
. htn_status derived.
. Total rows: 20442 | Yes: 8774 | No: 11655 | Missing: 13
. Prevalence: 42.9% (among non-missing rows)
. Source driving 'Yes' classification:
. HTA (measured BP, tier 1) : 8774
. antiHTA (documented Rx, tier 2) : 0
. crbpmed (self-reported, tier 3) : 0
. derive() complete.
. 408840 rows across 20 imputed datasets.
```